

HEALTHCARE ANALYTICS

Who Misses the Appointment?

Predicting patient no-shows so a clinic can act before the empty slot

Naga Sai Dintakurthi · 110,521 appointments · Python, scikit-learn

One in five appointments goes unused

20.2%

of 110,521 appointments were missed

≈ 22,300

empty slots another patient could have used

THE QUESTION

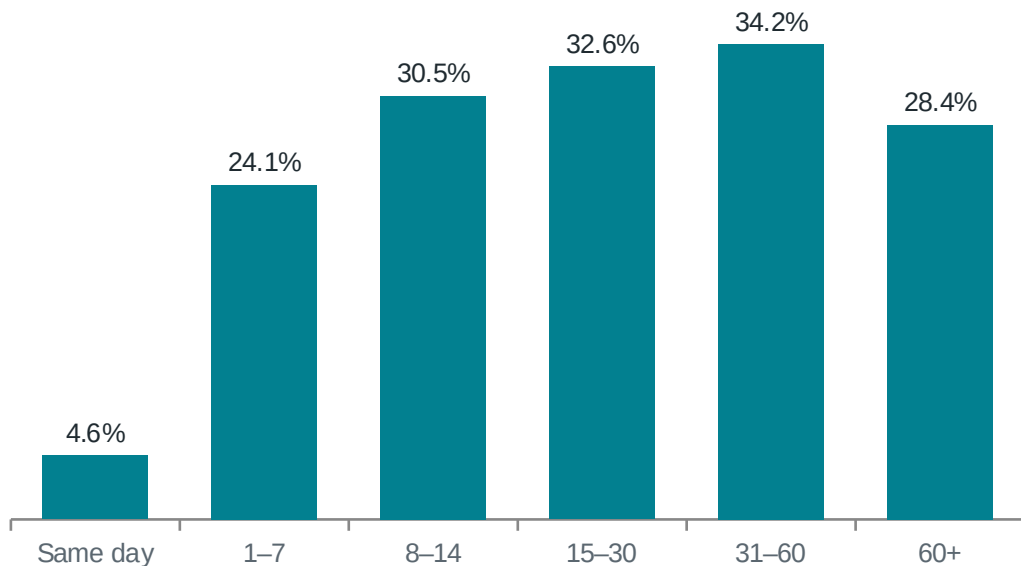
A clinic can't call every patient before every visit.

Which upcoming appointments are most likely to be missed, and what should the clinic do about them?

Data: public clinics in Vitória, Brazil (Kaggle, 2016).

Wait time is the strongest signal

No-show rate by days between booking and visit



TAKEAWAYS

Same-day visits: 4.6% no-show, and they're 35% of all bookings.

Past one week, ~1 in 3 patients miss.

The longer the gap, the more a reminder matters.

Teens and young adults miss the most

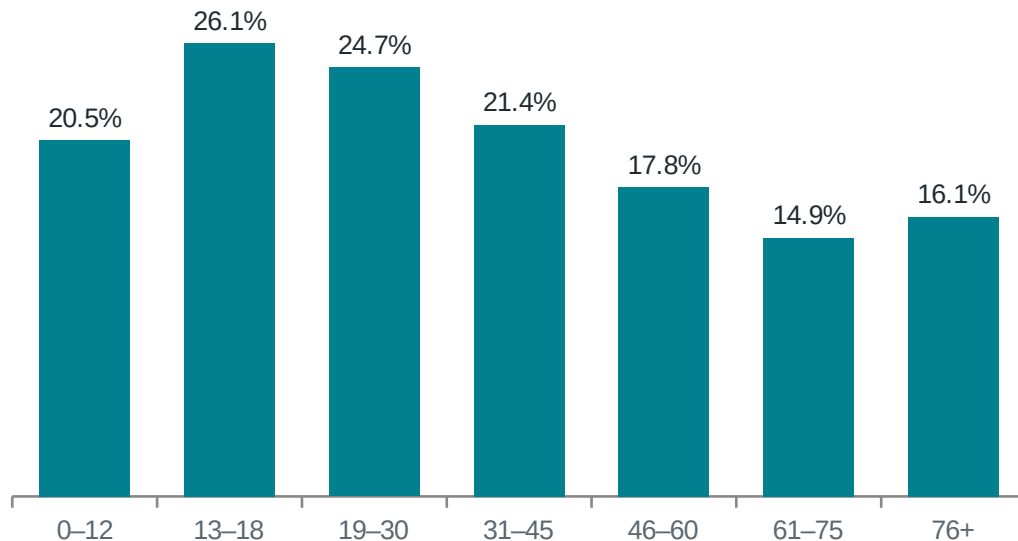
TAKEAWAYS

13–18: 26.1%, 19–30: 24.7%, the highest of any group.

Over 60s are most reliable (15–16%).

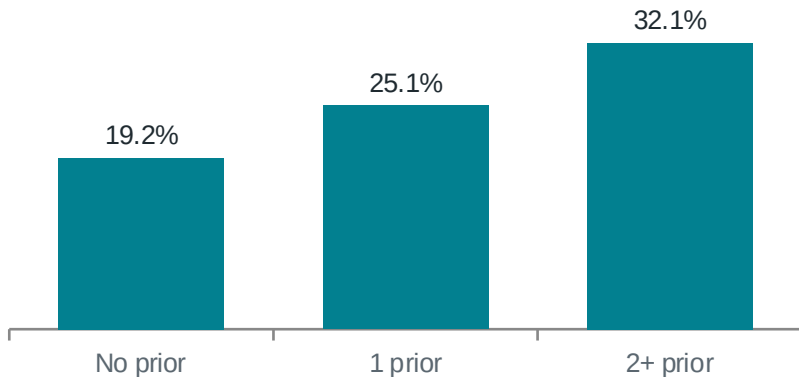
Remind under-30s by text; for minors, remind the parent.

No-show rate by patient age



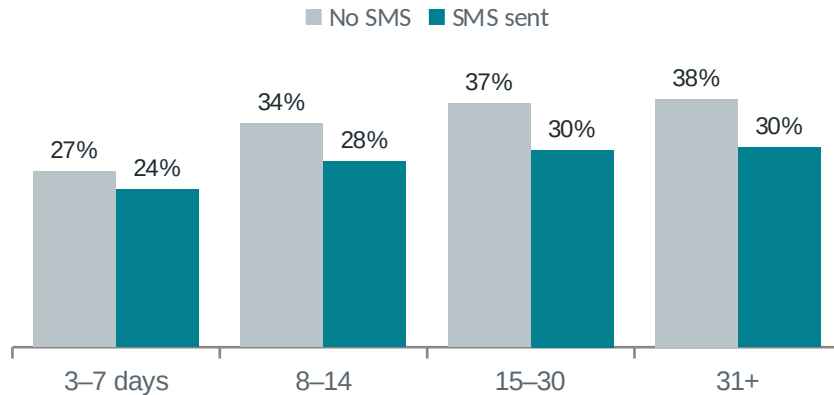
History predicts behavior, and SMS helps

No-show rate by earlier missed visits



2+ earlier no-shows → 32.1% vs 19.2% for a clean record. Flag at booking.

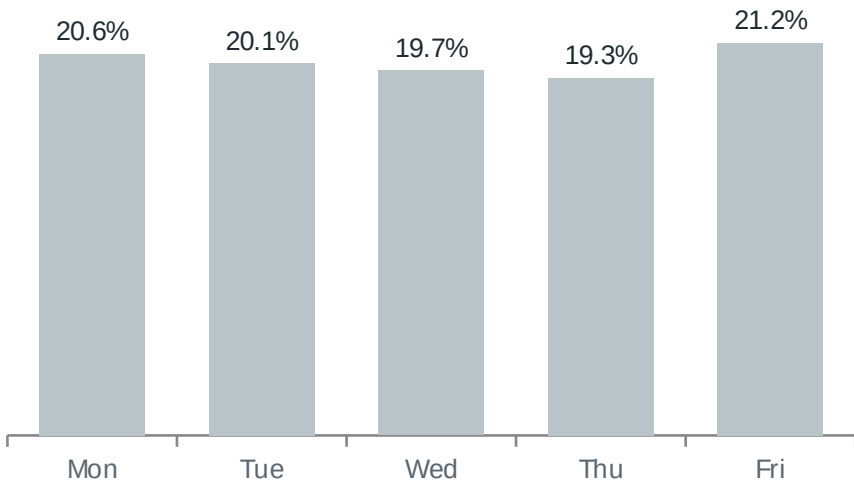
No-show rate at the same wait length



SMS lowers no-shows 3-7 pts at every wait length, yet 41% of eligible patients never got one.

What doesn't move the needle

No-show rate by weekday



Group	With	Without
Hypertension	17.3%	20.9%
Diabetes	18.0%	20.4%
Alcoholism	20.1%	20.2%

Weekday: Mon–Fri all within 19–21%. Rescheduling by day won't help.

Chronic conditions: patients miss the same or less. Don't target by diagnosis.

A risk score for every upcoming appointment

PURPOSE & PREPROCESSING

Purpose: rank patients so staff call the riskiest first.

Fixed a date bug that had dropped 38.5K same-day rows (35% of data).

Prior no-shows built from earlier visits only, so there's no look-ahead.

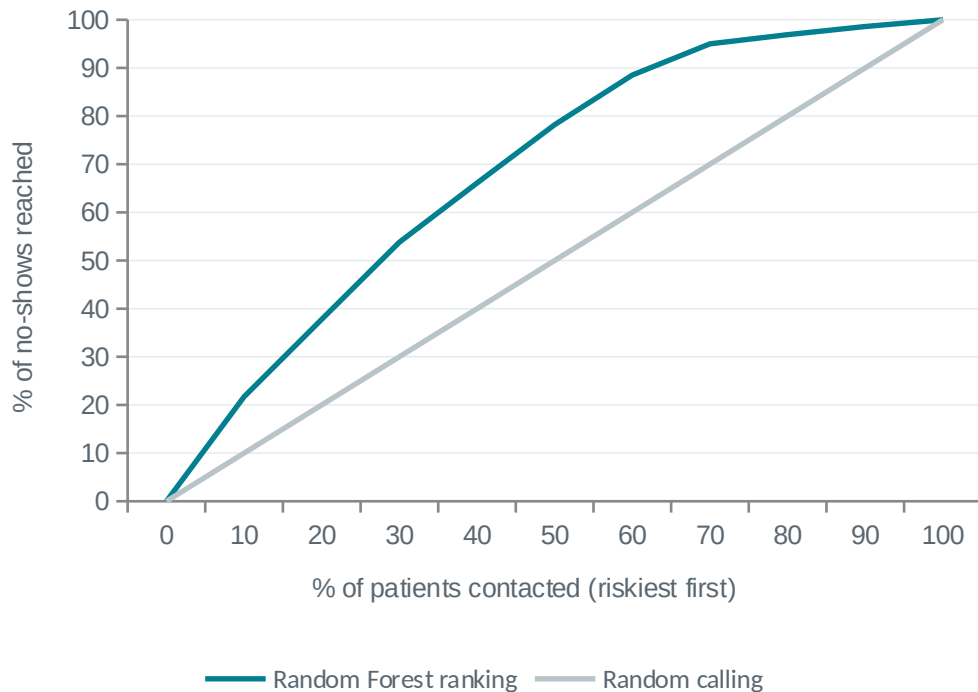
Not judged on accuracy: 'always show' already scores 79.8%.

Model	ROC-AUC	Recall	Precision
Logistic Regression	0.68	58%	32%
Decision Tree	0.73	81%	30%
Random Forest	0.74	82%	31%

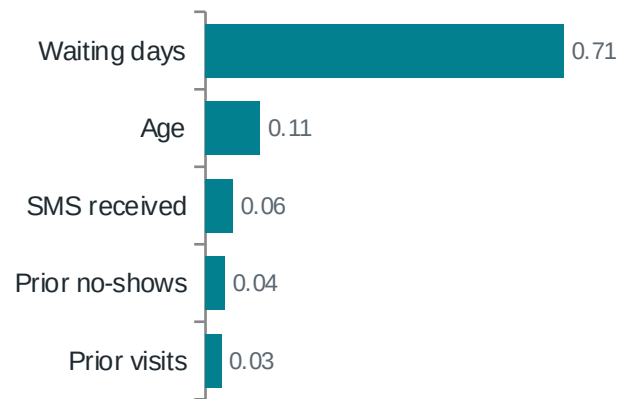
Recall and precision are for the no-show class, on a 20% held-out test set. Imbalance handled with balanced class weights.

Random Forest catches 8 in 10 no-shows.

Call 30% of patients, reach 54% of no-shows



WHAT DRIVES THE SCORE



The model learned the same patterns the EDA shows, led by wait time.

Five moves, ordered by impact

1

Score appointments nightly and call the top 30%

Reaches 54% of no-shows with 30% of the effort

2

Text every appointment booked 3+ days out

SMS cuts no-shows 3–7 pts; 41% of eligible patients get none

3

Open more same-day slots; confirm long waits

Same-day: 4.6% no-show vs 30%+ past one week

4

Flag under-30s and repeat no-shows at booking

Highest-risk groups (25–32%)

5

Stop targeting by weekday or diagnosis

Neither changes the no-show rate

Next steps

- Pilot recommendations 1 and 2 as an A/B test before a full rollout.
- Limitation: observational data from one city in 2016. The SMS effect is a correlation within wait bands, not a controlled experiment.
- Retrain on the clinic's own data. Add weather, distance and appointment type if available.

Code & data: github.com/nagasai-data/healthcare-no-show-dashboard